Emily E. Ackerman, Ph.D.

Postdoctoral Researcher in the Lahav Lab, Systems Biology, Harvard Medical School

Computational researcher with wide-ranging skills including single-cell sequencing analysis, statistics, mathematical modeling, and network biology. Experienced using python, R, and MATLAB for virology and cancer research applications. Committed to creating an equitable scientific enterprise for all.

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Education

August 2021 | Doctor of Philosophy in Chemical Engineering
University of Pittsburgh, Pittsburgh, PA | Advisor: Dr. Jason Shoemaker

May 2015 | Bachelor of Science in Chemical Engineering
Rensselaer Polytechnic Institute, Troy, NY

Research Experience

Aug 2021-Mar 2025 | Department of Systems Biology
Harvard Medical School | Dr. Galit Lahav | *Postdoctoral Research*

Developed an integrated **single-cell transcriptomic analysis** and **time series clustering** pipeline to correlate p53 protein dynamics and global transcription profiles within single cells. Discovered novel markers of heterogeneity in transcriptional programs of the MCF7 cell line post-DNA damage.

Uncovered correlations between p53 mutation types and comutated genes across four categories of myeloid neoplasia. Proved the utility of TP53 mutation identification in clinical diagnostics **(statistical analysis)**.

Discovered changing p53 dynamics with altered cell confluency is a function of Wip1 levels using **ODE modeling**.

Jan 2016-Aug 2021 | Department of Chemical and Petroleum Engineering
University of Pittsburgh | Dr. Jason Shoemaker | *Doctoral Research*

Identified host factors of influenza infection using virus-host protein network topology and controllability analyses. Evaluated **network methods** against high throughput biological screening methods.

Developed method to train multi-dataset, shared parameters for **ODE models** using Markov Chain Monte Carlo algorithm. Identified interferon production rate as the source of the strain-specific host immune response to influenza infection. Reviewed current models for sensitivity and ability to depict interferon pre-treatment.

Prioritized drug repositioning candidates for SARS-CoV-2 infection using **network controllability**. Participated in the international COVID-19 Disease Map effort to coalesce known molecular mechanisms of COVID-19.

Jun 2015-Aug 2015 | Intern at **Curia Global**
Computer-Aided Drug Discovery

Worked on a team of professionals towards the development of in-house **docking/scoring methods** for protein interactions. Optimized and automated all methods for department-wide use. Verbally presented results with all non-computational departments and management teams at end of term.

May 2013-May 2015 | Undergraduate Research Program
Rensselaer Polytechnic Institute | Dr. Curt Breneman | *Undergraduate Research*

Identified a library of potential microbicide ligands to inhibit HIV GP120-CD4 binding using high-throughput screening methods. Developed novel super-flexible **docking/scoring method** with binding site comparison in Autodock Vina and MOE. Assisted small team in writing an R21 NIH grant proposal.

Honors and Awards

Sept 2021-Sept 2023 | Research Supplement to Promote Diversity in Health-Related Research from the **National Institutes of Health (NIH)**

Honors and Awards Cont.

Apr 2021	Outstanding Research Assistant at the **University of Pittsburgh** *Awarded by the Engineering Graduate Student Organization*
Feb 2021	Dr. James Coull Memorial Fellowship Award for the **Department of Chemical Engineering, University of Pittsburgh** *Awarded annually to one senior Ph.D. student*
Dec 2019	Outstanding Ph.D. Paper, Summer 2019 for the **Department of Chemical Engineering, University of Pittsburgh** *"A Dual Controllability Analysis of Influenza Virus-Host Protein-Protein Interaction Networks for Antiviral Drug Target Discovery"*
Feb 2019	Chemical Engineering Department Research Day at the **University of Pittsburgh** *OXE Research Award, Best Oral Presentation* *"Network Methods for Identifying Regulators of Influenza A Virus"*
Sept 2018-Aug 2021	James H. Gilliam Fellowships for Advanced Study program at the **Howard Hughes Medical Institute** *Gilliam Fellow*

Peer-Reviewed Publications

Pub.	*Simerzin A.,* ***Ackerman E.****, Fujimaki K., Kohler R., Iwamoto Y., Weissleder R., Jambhekar A. & Lahav G. (2025)* "Cell Confluency Affects p53 Dynamics in Response to DNA Damage". *Molecular Biology of the Cell*
Pub.	*Jambhekar A.,* ***Ackerman E.****, Alpay B., Lahav G., & Lovitch S. (2024)* "Comparison of TP53 Mutations Across the Spectrum of Myeloid Malignancies Suggests Divergent Functional Roles in Initiation of Myelodysplasia and Progression to Acute Leukemia". *Blood Neoplasia*
Pub.	***Ackerman E.****, Weaver J., & Shoemaker J. (2022)* "Disparate Interferon Production Rate Drives Strain-Specific Immunodynamics of Influenza A Virus". *MDPI Viruses*
Pub.	*Bennett C.,* ***Ackerman E.****, Carrington P., & Fox S. (2020)* "Accessibility and The Crowded Sidewalk: Micromobility's Impact on Public Space". *Proceedings, 2021 ACM Designing Interactive Systems (DIS) conference*
Pub.	***Ackerman E.****, & Shoemaker J. (2020)* "Network Controllability-Based Prioritization of Candidates for SARS-CoV-2 Drug Repositioning". *MDPI Viruses*
Pub.	***Ackerman E.****, Mochan E., & Shoemaker J. (2019)* "Strain-Specific Immune Response to Influenza Virus Infection". *Part of special issue: 8th Conference on Foundations of Systems Biology in Engineering FOSBE 2019*
Pub.	***Ackerman E.****, Alcorn J., Hase T., & Shoemaker J. (2019)* "A Dual Controllability Analysis of Influenza Virus-Host Protein-Protein Interaction Networks for Antiviral Drug Target Discovery". *BMC Bioinformatics*

Peer-Reviewed Publications Cont.

Pub. | ***Ackerman E.**, Kawakami E., Katoh M., Watanabe, Watanabe T., Tomita Y., Lopes T., Matsuoka Y., Kitano H., Shoemaker J. & Kawaoka Y. (2018)* "Network-Guided Discovery of Influenza Virus Replication Host Factors". *mBio*

Pub. | ***Ackerman E.**, Mochan E., & Shoemaker J. (2018)* "A systems and treatment perspective of models of influenza virus-induced host responses". *MDPI Processes*

Select Research Presentations

Apr 2023 Co-pres. Talk | "Distinct TP53 mutation spectra in myeloid neoplasms suggest divergent roles in disease initiation and progression"
Ludwig Center Weekly Meeting, **Harvard Medical School**

Nov 2022 Poster | "Comparison of TP53 Mutations Across the Spectrum of Myeloid Malignancies Suggests Divergent Functional Roles in Initiation of Myelodysplasia and" Progression to Acute Leukemia"
Ludwig Center Annual Meeting, **Harvard Medical School**

Aug 2022 Poster | "Time Series Clustering for the Integration of p53 Protein Dynamics and Transcriptomics in Single Cells"
NIH Diversity Supplement Professional Development and Networking Workshop, **National Institutes of Health (NIH)**

Nov 2021 Invited Talk | "Controls Engineering Approaches to Regulating Immunity During Respiratory Infection"
U-RISE Seminar Speaker, **University of Maryland, Baltimore County**

Oct 2021 Talk | "Interferon Production Rate is a Major Contributor to Differential Strain-Specific Immunodynamics"
5th Workshop on Virus Dynamics, **Fred Hutchinson Cancer Research Center**

Jul 2020 Talk | "Identifying Regulators of Infection in Virus-Host Networks"
International Conference on Intelligent Systems for Molecular Biology, ISMB, **Virtual**

Feb 2019 Talk | "Network Methods for Identifying Regulators of Influenza A Virus Infection
Chemical Engineering Department Research Day, **Pittsburgh, PA**

Oct 2018 Invited Talk | "Controllability of the Influenza Virus-Host Protein-Protein Interaction Network: Engineering Insights into Host-Virus Interactions
American Institute of Chemical Engineers, Annual Meeting, **Pittsburgh, PA**
Area Plenary: Future Directions in Applied Mathematics and Numerical Analysis

Teaching Experience

Fall 2016-2018 | Teaching Assistant at the **University of Pittsburgh**
Systems Engineering 1: Dynamics and Modeling | Dr. Jason Shoemaker
Taught recitation for senior undergraduates including new concepts and practice problems. Planned and taught guided simulations in MATLAB and Simulink. Provided examples of challenging material after skill assessments.

Professional Leadership and Service

Dec 2021-Mar 2025 | Disability Advisor for the **Department of Systems Biology** at **Harvard Medical School**

Provided guidance to department leadership on accessibility of physical space, websites, etc. Acted as liaison between students, researchers, staff, and department administration concerning disability issues. Established virtual community for disabled researchers within Harvard Medical School. Created new onboarding materials regarding procedures for making accessibility improvements in lab spaces.

Aug 2020-Feb 2024 | Board of Directors of **Future of Research**

Co-led the Labor Task Force for the investigation of graduate student and postdoc labor issues. Conceived and carried out large scale survey of workplace conditions for academic early career researchers. Worked with Board of Directors and Executive Board to empower junior researchers through equitable, grassroots action.

Jan 2023-Jun 2023 | Committee Member for **Beyond Compliance: Promoting the Success of People with Disabilities in the STEM Workforce** at the **National Academies of Sciences, Engineering, and Medicine (NASEM)**

Organized and hosted five day workshop to explore issues related to the accessibility and inclusivity of STEM workplaces. Created lasting resources in the form of video recordings and conference proceedings.

Dec 2022-Dec 2023 | Member of the **Advisory Committee to the Director - Working Group on Re-envisioning NIH-Supported Postdoctoral Training** at the **National Institutes of Health (NIH)**

Engaged with key parties to evaluate factors contributing to the declining number of U.S. postdoctoral researchers. Reported evidence and wrote recommendations to optimize both the postdoctoral training experience and the scientific enterprise overall.

Sept 2021-Jun 2022 | Committee Member for the **Committee on Leading Practices for Improving Accessibility and Inclusion in Field, Laboratory, and Computational Science** at the **National Academies of Sciences, Engineering, and Medicine (NASEM)**

Organized and hosted five webinars including keynotes and guided Q&A with disabled speakers to discuss the current state STEM research for disabled researchers and provide recommendations for the future. Created lasting resources in the form of video recordings.

May 2021 | Organizer of the **Valuing Disabled Voices in STEM Workshop** at the **University of Pittsburgh**

Conceived and organized a workshop to highlight the experiences and research of disabled faculty and students.

Jan 2017-May 2020 | President of Graduate Women Engineering Network
University of Pittsburgh

Prepared workshops on skills and topics which benefit members such as pay negotiation, navigating impostor syndrome, and Title IX panels. Organized social events and peer mentoring groups for women in STEM to network. Planned and lead general body meetings and executive board meetings.

Select Invited Lectures and Panels on Equity

April 2024 Talk | "Diversity - Inclusion and Accessibility"
Research and Application in Team Science Committee, **National Academies of Sciences, Engineering, and Medicine (NASEM)**

April 2024 Panel | "Becoming a Researcher: Inclusion and Accessibility in the Lab Environment"
Inclusion, Diversity, Equity, and Antiracism Series (IDEAS), **Association of American Medical Colleges (AAMC)**

Select Invited Lectures and Panels on Equity Cont.

Jan 2024 Panel	"Roles, Responsibilities, and Expectations of Graduate Students and Postdoctoral Scholars: The Current Landscape and History of Graduate Student and Postdoctoral Scholar Labor Movements" *Roundtable on Mentorship, Well-being, and Professional Development,* **National Academies of Sciences, Engineering, and Medicine (NASEM)**
July 2023 Talk	"Creating an Anti-Ableist Future for Science" *DEI Speaker Series,* **National Academies of Sciences, Engineering, and Medicine (NASEM)**
Jan 2023 Talk	"Accessibility and Inclusion Conversation Series" *Distinguished Lecture,* **National Science Foundation**
Sept 2020 Talk	"The Disability and Tech Accessibility Cycle" *Pitt Grad Student Organizing Committee, STEM and Society Lecture Series,* **University of Pittsburgh**
Apr 2020 Talk	"The Accessibility Gap for Tech Users and Developers" *Carnegie Mellon University, Accessibility Group,* **Pittsburgh, PA**

Skills

Techniques:	Genomics analysis *(Seurat, DESeq2, IPA)*, timecourse clustering, mathematical modeling, Markov Chain Monte Carlo for model parameterization, network analysis, statistics
Programming:	R *(Bioconductor, tidyverse, ggplot2)*, Python *(Pandas, NumPy, SciPy, Scikit-learn, matplotlib, seaborn)*, Matlab, Jupyter Notebook, Git, Bash, LaTeX
General:	Excel, Word, PowerPoint, research design, written and oral communication, leadership, mentorship, self-directed problem solving